

Reading Time: An initial 2 minutes to view **BOTH** sections



MATHEMATICS METHODS : UNITS 1 & 2, 2023

PM

Test 4 – (10%)

2.2.4-2.2.9, 2.3.2-2.3.17, 2.3.20-2.3.21

Time Allowed 20 minutes	First Name	Surname	Marks
			23 marks

Circle your Teacher's Name:

Mrs Alvaro	Mr Bartoshefski	Mr Buckland	Mr Liu
Mr Lockyer	Mr Luzuk	Mr Mohammadi	
Ms Narendranathan		Mr Ng	
Mr Stephens	Mr Tanday		

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Not Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

PART A – CALCULATOR FREE

Question 1 (3 marks)

For an arithmetic series, the first term is 10 and the sum of the first 2 terms is 16.

Show that the common difference is -4.

Question 2 (1, 2, 3 = 6 marks)

- (a) Determine $\frac{dy}{dx}$ for the following, leaving your answer in simplest form, with positive indices:

(i) $y = 3x^4 - 2x + 5$

(ii) $y = \sqrt[3]{x} - 5x$

- (b) Find the value of the gradient of $y = \frac{8x^4 - 3x^2}{x^3}$ at $x = 3$.

Question 3 (4 marks)

Differentiate $f(x) = 2x^2 - x$ by first principles.

Question 4 (5, 3, 2 = 10 marks)

A function has the following rule $y = 2x^3 - 6x^2 + 1$.

- (a) Determine, using calculus, all the stationary points and their nature, using the sign test, for the function given above.
- (b) Determine the global maxima and minima over $-1 \leq x \leq 4$.
- (c) Discuss the behaviour of the function as $x \rightarrow \mp\infty$.

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PM

Test 4 – (10%)

2.2.4-2.2.9, 2.3.1-2.3.17, 2.3.20-2.3.21

Time Allowed	First Name	Surname	Marks
30 minutes			32 marks

Circle your Teacher's Name:

Mrs Alvaro	Mr Bartoshefski	Mr Buckland	Mr Liu
Mr Lockyer	Mr Luzuk	Mr Mohammadi	
Ms Narendranathan		Mr Ng	
Mr Stephens	Mr Tanday		

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

PART B – CALCULATOR ASSUMED

Question 5 (5, 2 = 7 marks)

If $(t + 1)$, $(1 - t)$ and $(2 - 5t)$ are the first three terms of an infinite, **converging** geometric sequence, determine:

(a) the value(s) of t and the common ratio.

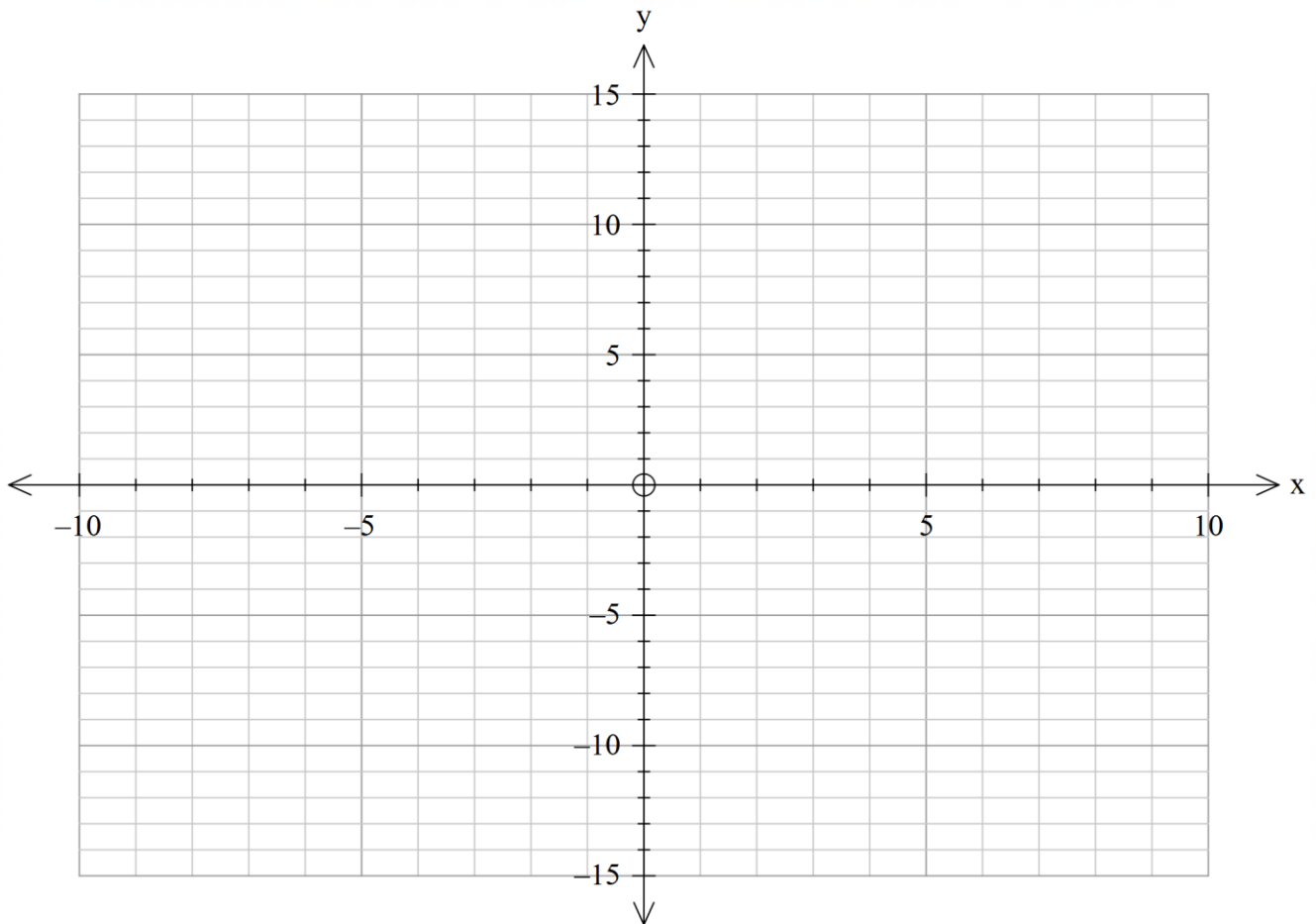
(b) the sum to infinity.

Question 6 (4 marks)

A function has the following characteristics:

- The function has stationary points are $x = -3$, $x = 2$, and $x = 7$.
- The function has roots at $x = -5$, $x = -1$, $x = 5$ and $x = 9$
- The function has a local maximum at $x = 2$ and local minimums at $x = 7$ and $x = -3$.
- $f'(x) > 0$ for $-3 < x < 2$ and $x > 7$.
- $f'(x) < 0$ for $x < -3$ and $2 < x < 7$.
- The function has a y-intercept at $(0, 6)$.
- The function changes concavity at $x = 0$ and $x = 5$.

Graph an example of the function on the axes provided below.



Question 7 (2, 2 = 4 marks)

Determine the coordinates on the graph of $f(x) = x^3 - 3x^2 + 3x + 4$ where the tangent:

(a) has a gradient of 12.

(b) is parallel to the line $3x - y = -4$.

Question 8 (2, 2, 1, 3, 3 = 11 marks)

Indy accepts employment with a first year salary of \$52 000. In each of the following years her annual salary is increased by \$2575 per year.

Skye accepts employment with a first year salary of \$52 000. In each of the following years her annual salary is increased by 4%.

- (a) State the recursive formula for Indy's annual salary.

- (b) State the general (explicit) formula for Skye's annual salary during the n^{th} year.

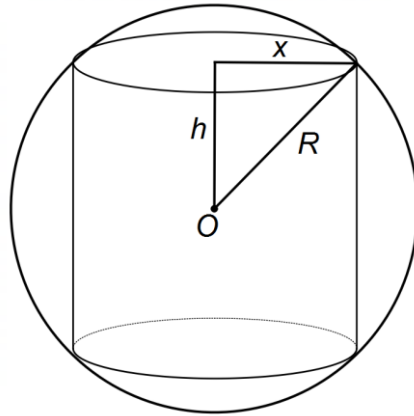
- (c) Determine Skye's annual salary during the 10th year.

- (d) During which year does Skye earn more than Indy for the first time? Show evidence to justify your answer.

- (e) By what amount does the total amount earned by Skye exceed that earned by Indy at the end of the twentieth year?

Question 9 (3, 3 = 6 marks)

A cylinder, of radius x metres and height $2h$ metres, is to be inscribed in a sphere of **constant radius R (m)** centred at O as shown below.



(a) Show that the volume of the cylinder is given by $V = 2\pi h(R^2 - h^2)$.

(b) Use **Calculus techniques** to show that the cylinder has a maximum volume when $h = \frac{\sqrt{3}R}{3}$. You **do not** need to use the sign test to justify.

